Indian Two-Wheeler Market Analysis

End-to-end web-scraping project using BikeWale.com data (446 models)

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Business problem and objectives

A data analyst at a two-wheeler manufacturer must show management where the market is, what customers rate highly and where electric vehicles stand.

Market structure

Petrol vs electric, scooter vs motorcycle, brands

Price and segments

Which price bands and segments dominate

Satisfaction

Does a higher price buy higher ratings?

Electric vs petrol

Price, range and rating compared like-for-like

Data collection: web scraping

Website

BikeWale “Best Bikes in India” list, sorted by popularity

Scraper

Python, requests, BeautifulSoup, regex

Raw JSON

446 models × 13 fields

pandas

Cleaning, features, analysis

Outputs

Cleaned CSV, dictionary, cleaning log, charts

Polite scraping: 2-second delay, retries and a browser user-agent; only public data, for educational use. The loop stops automatically at the first empty page (15 pages with data).

Data cleaning: key decisions

446 rows in, 446 out · 0 duplicates · 18 columns after feature engineering

Missing values kept

EV-only fields are ~76% empty for petrol bikes; left as NaN, never imputed.

Unrated models

35 models (7.8%) have no rating; rating analysis uses rated models only.

Reliable ratings

Ratings from fewer than 10 reviews are excluded from rating analysis.

New features

Brand, Fuel_Type, Price_Lakh, Price_Band and Segment.

Segment column

Superbikes up to ₹100 lakh sit beside ₹1 lakh commuters, so comparisons use segments.

Outliers

39 price outliers are genuine superbikes: kept, medians used.

EDA: four segments, two very different markets

109

Electric

Median price ₹1.12 L · median rating 4.0

110

Commuter petrol (up to 200cc)

Median price ₹1.16 L · median rating 4.5

103

Mid petrol (201–500cc)

Median price ₹2.35 L · median rating 4.6

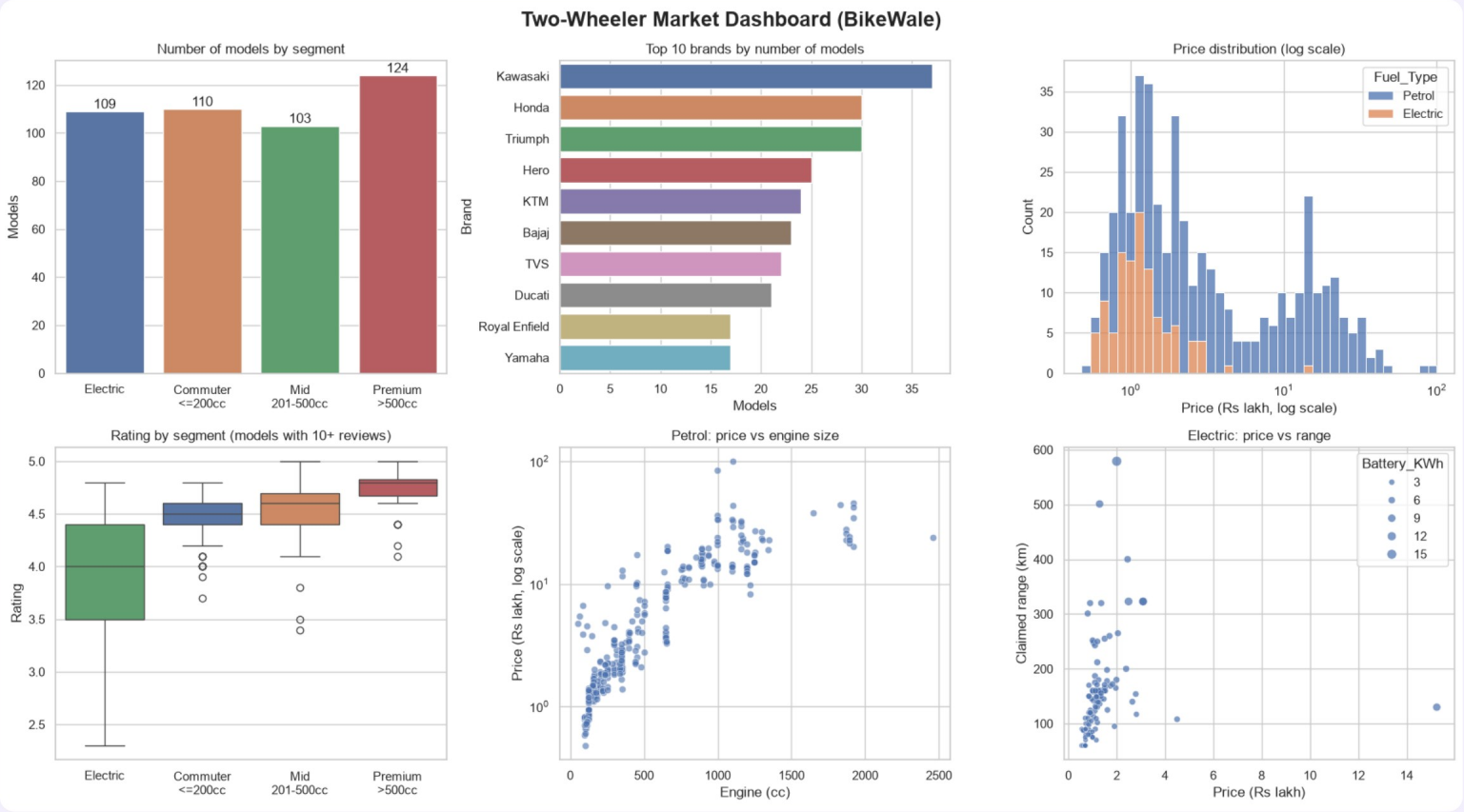
124

Premium petrol (above 500cc)

Median price ₹15.02 L · median rating 4.8

58 brands. Kawasaki lists the most models (37), then Honda and Triumph (30 each). Price runs from ₹0.5 to ₹100 lakh, so overall averages mislead.

Dashboard



Price needs a log scale: ₹0.5–100 lakh.

EV ratings sit below every petrol segment.

Price rises steeply with engine size.

Insights: price and satisfaction

−3%

EV vs commuter petrol price

Median ₹1.12 L vs ₹1.16 L: electric is price-competitive

0.5 ★

EV rating gap

Median rating 4.0 (EV) vs 4.5 (commuter petrol)

0.51

Price vs rating

Overall Spearman correlation; inside segments it still averages 0.26, so higher-priced models tend to be rated higher

Electric two-wheelers cost the same as petrol commuters but are rated lower by owners. Median ratings rise from 4.0 (EV) to 4.5 (commuter), 4.6 (mid) and 4.8 (premium).

Insights: value leaders and brands

Best-value EVs (₹ per km of range)

OLA Roadster X Plus, OLA S1 Z, OLA S1 X

Best mileage (commuter petrol)

TVS Sport, Hero Super Splendor Xtec 2.0, Bajaj Platina 100

Highest-rated everyday brands

KTM 4.60, Yamaha 4.52, Hero 4.47 (EV + commuter petrol; 3+ models, 10+ reviews)

Everyday market (EV + commuter petrol): the most crowded price band is ₹1–1.5 lakh (41% of these models).

Recommendations

1. Plan, price and benchmark each segment separately; never compare a commuter with a superbike.
2. Close the EV satisfaction gap: EVs are priced like petrol commuters but rated 0.5 stars lower, so invest in reliability, software and service first.
3. Price new EVs at or below the cost per km of range of OLA Roadster X Plus.
4. In the crowded ₹1–1.5 lakh band, differentiate on features and service, or look for gaps in neighbouring bands.
5. Learn quality and service practices from the best-rated everyday brands (KTM, Yamaha).
6. Run review campaigns for new launches: ratings under 10 reviews are unreliable.
7. Re-run the scraper monthly to track prices and EV adoption.

Limitations, next steps and conclusion

Limitations

- One website, one date
- Listed price, not on-road price
- Popularity is not sales
- Ratings are self-selected; range and mileage are claimed

Next steps

- Scrape detail pages for variants and review text
- Monthly price trends
- Regression model to find over- and under-priced models

A complete pipeline: scrape → clean → analyse → visualise → recommend. Thank you · Questions?